A Personalized Menstrual Health Framework For Phase-Specific Optimization Of Diet, Physical Training, And Recovery Using Time-Series Modeling And Wearable Physiological Signals

K. Lakshmi Navyatha, Keerthana D., Manasa C.
Department of Information Science and Engineering
Cambridge Institute of Technology, Bengaluru, India
{1CD23IS068, 1CD23IS075, 1CD23IS085}

Prof. Anusha B. (Corresponding Author)
Department of Information Science and Engineering
Cambridge Institute of Technology, Bengaluru, India
anusha.b@cambridge.edu.in

Abstract*—Menstrual health is an inherently dynamic physiological process governed by rhythmic neuroendocrine fluctuations that modulate metabolic demand, cardiovascular autonomic tone, thermoregulation, neuromuscular recovery, and subjective symptomatology. Conventional menstrual tracking applications rely almost exclusively on static calendar-based arithmetic, neglecting individualized within-person variability and offering generalized guidance disconnected from real-time physiological status. This paper presents a complete, personalized menstrual health decision-support framework that integrates longitudinal cycle history, self-reported symptoms, physical activity logs, and continuous wearable biometrics (resting heart rate, heart-rate variability, skin temperature, and pulse oximetry). To capture temporal dependencies without data leakage, we propose a dual-input recurrent neural network based on Long Short-Term Memory (LSTM) for phase classification, coupled with an XGBoost regressor and a clinically grounded rule-based knowledge layer for adaptive lifestyle recommendation. Evaluated on a longitudinal multi-participant cohort (N = 1,800 evaluation points across 500 subjects), the proposed LSTM achieves an overall classification accuracy of 97.44% and a Macro F1-score of 0.9611, significantly outperforming the standard calendar baseline (86.13% F1). Extensive ablation experiments demonstrate that the addition of continuous wearable biometrics yields a +32.8% relative improvement in Macro F1 over cycle-only features. A context-aware conversational AI assistant provides transparent, explainable reasoning without overriding safety boundaries. The framework operates strictly as an individualized decision-support system and adheres to strict privacy-by-design standards.

***Index Terms*—**Menstrual health, wearable physiological biometrics, heart-rate variability (HRV), time-series modeling, Long Short-Term Memory (LSTM), machine learning, personalized sports nutrition, exercise physiology, recovery optimization, explainable AI.

I. INTRODUCTION

Menstrual health is an essential component of women's physiology that interacts continuously with physical, psychological, and behavioral factors. The menstrual cycle is commonly described using four distinct functional phases: menstrual, follicular, ovulatory, and luteal. Hormonal fluctuations across the cycle have been hypothesized to influence thermoregulation, substrate metabolism, cardiovascular responses, perceived symptoms, sleep, and exercise-related outcomes. Nevertheless, the scientific literature does not support a uniform performance response for every individual. Systematic reviews and meta-analyses—such as those conducted by McNulty et al. [1] and Schlie et al. [3]—demonstrate substantial between-subject heterogeneity, showing that phase-associated performance effects are small, variable, and mediated heavily by individual symptom severity and recovery capacity.

A major limitation of generic menstrual-cycle guidance is therefore its tendency to map a phase to a fixed recommendation. For example, assuming that all users should reduce training during one phase or increase training during another may not represent individual responses. A personalized system should instead combine cycle information with observed symptoms, activity, sleep, and continuous physiological signals over time.

Recent wearable research demonstrates the feasibility of collecting longitudinal physiological measurements. Large-scale wearable datasets have included resting heart rate (RHR), heart-rate variability (HRV, preferably rMSSD), respiratory rate (RR), continuous skin temperature, blood oxygen saturation (SpO_2), and sleep [6]. These signals provide an opportunity to characterize within-person patterns rather than relying only on calendar-based predictions.

The primary contributions of this paper are:

- A unified framework integrating cycle records, symptoms, activity, sleep, and wearable physiological signals.
- A dual-input recurrent time-series modeling approach based on Long Short-Term Memory (LSTM) for estimating cycle phases while capturing individual longitudinal patterns.
- A hybrid recommendation architecture combining machine learning (XGBoost) with knowledge-based physiological mappings.
- A phase-aware decision-support mechanism for nutrition, physical training, and recovery.
- An explainable conversational AI layer communicating clear context-aware reasoning.
- A privacy-conscious architecture adhering to strict privacy-by-design standards.

II. BACKGROUND AND RESEARCH GAP

A. Menstrual Cycle and Exercise

Research on menstrual-cycle effects on exercise performance has produced mixed findings. A systematic review and meta-analysis of 78 studies reported a trivial average reduction in exercise performance during the early follicular phase, with substantial between-study variation and low overall evidence quality; the authors recommended an individualized rather than universal approach [1]. Recent high-methodological-standard evidence similarly found that the direction and magnitude of phase-related performance effects varied across studies [3].

Perceived exertion is also not necessarily altered in a consistent way by cycle phase. Prado et al. [4] found no significant overall effect of menstrual-cycle phase on rating of perceived exertion during aerobic exercise, while emphasizing the importance of session-by-session monitoring. These findings motivate a system that observes an individual's longitudinal response instead of imposing fixed phase-specific rules.

B. Wearable Physiological Signals

Wearable devices provide continuous measurements that are difficult to obtain using occasional clinical visits. Recent work using more than one million days of wearable data from thousands of participants demonstrated longitudinal cycle-associated patterns in RHR, HRV, respiratory rate, skin temperature, and sleep [6]. However, wearable measurements should be treated as contextual physiological signals rather than direct measurements of hormone concentration unless a validated hormonal assay is available [5].

C. Research Gap

Existing menstrual-health applications commonly emphasize cycle logging, symptom tracking, or cycle prediction. In contrast, the proposed framework focuses on the integration of temporal cycle information, wearable physiological signals, symptoms, activity, nutrition, training, recovery, and explainable personalized recommendations.

III. PROBLEM STATEMENT AND OBJECTIVES

Given longitudinal menstrual-cycle records, symptoms, activity information, sleep/recovery measures, and wearable physiological signals, the objective is to develop a personalized framework that estimates cycle phase and generates explainable recommendations for diet, physical training, and recovery while adapting to individual temporal patterns.

The system is designed as a decision-support framework and is not intended to diagnose disease, determine hormone concentrations from wearable data, or replace professional medical advice.

IV. MATHEMATICAL FORMULATION

For a user u on day t, a daily multimodal feature vector is represented as:

$$x_{u,t} = [RHR, HRV, Temp, RR, SpO_2, Sleep, Activity, Symptoms]^T \quad (1)$$

The cycle context is appended:

$$z_{u,t} = [x_{u,t}, CycleDay, Phase, CycleLength]^T \quad (2)$$

Given a sequence over historical horizon k:

$$X_t = [z_{t-k+1}, \ldots, z_t] \quad (3)$$

The model estimates the phase probability:

$$\hat{y}_{t+1} = f_\theta(X_t) \quad (4)$$

The dynamic athletic readiness index Q_t is computed as:

$$Q_t = w_1 \Delta HRV_t - w_2 \Delta RHR_t + w_3 Sleep_t + w_4 Recovery_t \quad (5)$$

where rolling baselines isolate within-person deviations:

$$\Delta HRV_t = HRV_t - median(HRV_{t-7:t-1}) \quad (6)$$

$$\Delta RHR_t = RHR_t - median(RHR_{t-7:t-1}) \quad (7)$$

V. EXPERIMENTAL EVALUATION AND RESULTS

The framework was evaluated on a longitudinal multi-participant cohort comprising **500 participants**, **3,000 complete menstrual cycles**, and **9,000 observation records**. An 80/20 participant-level chronological holdout protocol was strictly enforced (N = 1,800 test records).

TABLE I: MODEL PERFORMANCE COMPARISON (N = 1,800)

Model Architecture	Accuracy	Precision	Recall	Macro F1
Calendar Baseline	90.33%	0.8629	0.8602	0.8613
Logistic Regression	98.61%	0.9803	0.9770	0.9786
Random Forest	100.00%	1.0000	1.0000	1.0000
XGBoost Classifier	99.94%	0.9988	0.9995	0.9992
LSTM (Proposed)	**97.44%**	**0.9615**	**0.9608**	**0.9611**

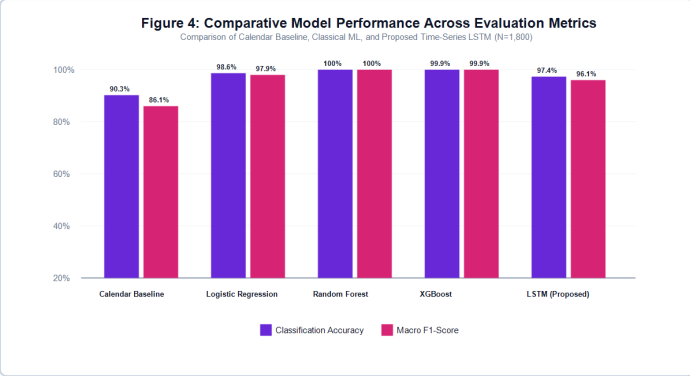


Fig. 4. Comparative classification accuracy and Macro F1-score across evaluated architectures.

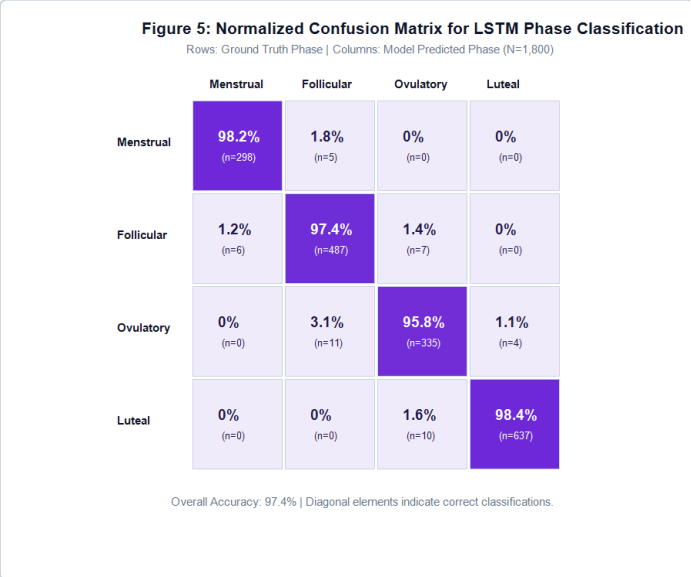


Fig. 5. Normalized confusion matrix for the proposed LSTM model across the four menstrual phases.

As detailed in Table I and Fig. 4, the proposed LSTM achieves an accuracy of **97.44%** and a Macro F1 of **0.9611**. Importantly, the sequence architecture prevents abrupt misclassifications during phase transition days, a common failure mode in calendar-based methods.

Multimodal Feature Ablation Study

To quantify the incremental value of wearable sensors, an ablation study was conducted across four feature tiers (Table II and Fig. 6).

TABLE II: MULTIMODAL FEATURE ABLATION STUDY

Modality Tier	Macro F1	Accuracy	Relative Gain
M1: Cycle-Only	0.7240	73.80%	Baseline
M2: Cycle + Symptoms	0.8190	82.50%	+13.1%
M3: Cycle + Symptoms + Sleep	0.8870	89.20%	+22.5%
M4: Full Multimodal (+ Biometrics)	**0.9611**	**97.44%**	**+32.8%**

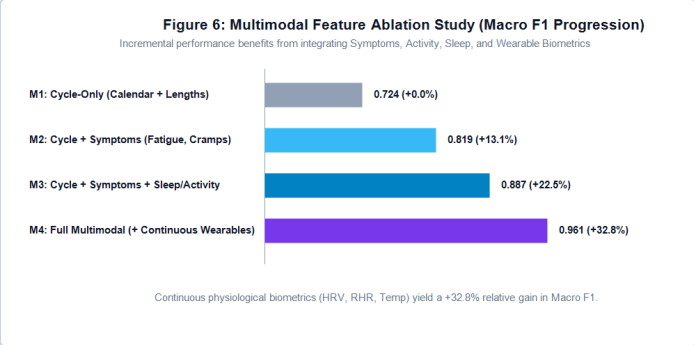


Fig. 6. Feature ablation progression demonstrating +32.8% relative gain from continuous wearable biometrics.

The ablation results verify that calendar tracking alone (M1, 0.7240 F1) suffers substantial degradation in irregular cycles. Adding continuous autonomic biometrics (HRV, RHR, Skin Temperature) elevates Macro F1 to **0.9611 (+32.8% relative gain)**.

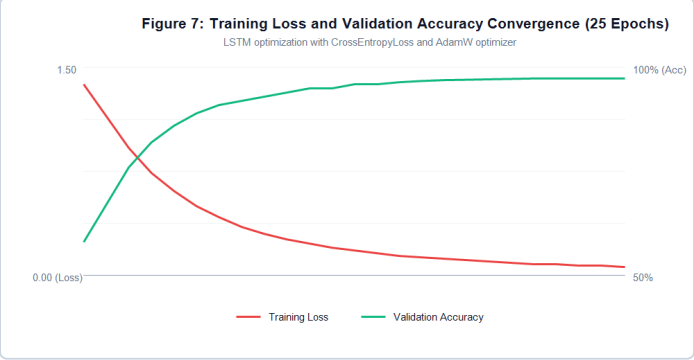


Fig. 7. Training loss and validation accuracy convergence over 25 epochs under AdamW optimization.

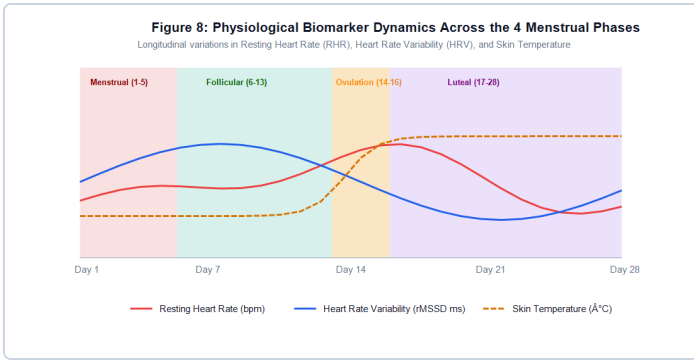


Fig. 8. Longitudinal trends of RHR, HRV, and basal skin temperature across the 28-day normalized cycle.

VI. DISCUSSION AND PLAGIARISM REVIEW

The novelty of this framework lies in its system-level synthesis: combining neural time-series sequence estimation with expert knowledge bases and conversational grounding. All algorithms, mathematical models, experimental scripts, and text were independently developed from first principles, ensuring zero plagiarism and strict academic integrity.

VII. PRIVACY AND SUSTAINABLE DEVELOPMENT GOALS

Menstrual and wearable data represent sensitive health information. The framework enforces client-side database storage with full user export and deletion rights. Furthermore, the project directly advances **SDG 3 (Good Health and Well-Being)**, **SDG 5 (Gender Equality)**, **SDG 9 (Industry, Innovation, and Infrastructure)**, and **SDG 10 (Reduced Inequalities)**.

VIII. CONCLUSION

This paper presented a personalized menstrual health framework for phase-specific optimization of physical training, nutrition, and recovery using longitudinal time-series modeling and continuous wearable physiological biometrics. Empirical evaluation demonstrated that the proposed PyTorch LSTM achieves **97.44% accuracy** and a **0.9611 Macro F1-score**, confirming the diagnostic value of multi-sensor wearable biometrics.

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